



Harshitha S 250701234-2025-CSE ▾

H2**Started on** Monday, 5 October 2026, 4:08 PM**State** Finished**Completed on** Monday, 5 October 2026, 4:09 PM**Time taken** 30 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n;
5      scanf("%d", &n);
6
7      int a[n];
8
9      for (int i = 0; i < n; i++) {
10         scanf("%d", &a[i]);
11     }
12
13     int k;
14     scanf("%d", &k);
15
16     int i = 0;
17     int j = 1;
18
19     while (i < n && j < n) {
20
21         if (i != j && a[j] - a[i] == k) {
22             printf("1");
23             return 0;
24         }
25
26         if (a[j] - a[i] < k) {
27             j++;
28         }
29         else {
30             i++;
31         }
32
33         if (i == j) {
34             j++;
35         }
36     }
37
38     printf("0");
39

```

```
40 |         return 0;  
41 |     }
```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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